

# Phase-Synchronous Distributed Regulation: A Unified Theory of Oscillator-Referenced Cell-Partition Control

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## Abstract

Classical control theory parametrises plant dynamics in physical time with dimensional state vectors; cross-domain composition (temperature  $\times$  pressure) requires explicit unit bookkeeping and yields no canonical convergence proof for distributed heterogeneous systems. This paper develops a phase-synchronous distributed control theory in which (i) every physical channel is normalised to the unit interval  $[0, 100]$  via  $S$ -entropy coordinates, (ii) oscillator phase  $\theta$  replaces physical time as the independent variable, giving a dimensionless phase-domain transfer matrix  $\mathbf{G}(\nu)$  valid across physical domains, (iii) the timing deviation  $\Delta P(k) = T_{\text{ref}}(k) - t_{\text{rec}}(k)$  serves simultaneously as measurement primitive and process output, and (iv) a finite cell partition of the normalised state hypercube defines a quantised sampled-data controller whose stability is certified by a piecewise Lyapunov function. We prove ten principal results: the Phase-Domain Equivalence theorem, Piecewise Stability theorem, Phase-Lock Lyapunov Correspondence, CUSUM- $\Delta P$  Identity, Multi-Scale Hierarchy Stability, Banach Contraction convergence, Delay Immunity bound, Temporal Nyquist criterion, Information-Optimal Sensor Selection, and Structural Incorruptibility. Together these constitute a self-contained control theory for large-scale distributed cyber-physical systems.

**Keywords:** distributed control, phase synchronisation,  $S$ -entropy normalisation, cell-partition quantisation, piecewise Lyapunov stability, Kuramoto oscillators, CUSUM duality, cyber-physical security.

## 1 Introduction

Large-scale cyber-physical systems (CPS) routinely integrate sensors and actuators from incommensurable physical domains: temperatures measured in kelvins, pressures in pascals, flow rates in cubic metres per second. Classical linear control theory elegantly handles each channel independently, but distributed multi-channel architectures expose a fundamental dimensional incompatibility problem: the product or ratio of transfer functions from heterogeneous channels carries mixed units, so cross-channel

composition demands explicit bookkeeping at every algebraic step. This bookkeeping burden is not merely notational; it propagates into numerical instabilities when dimensional ratios span many orders of magnitude and into logical errors when engineers omit conversion factors under time pressure [?].

A second difficulty arises from the choice of independent variable. Physical-time parametrisation forces every subsystem to share a common clock, an assumption that is poorly satisfied in wireless sensor networks, power-line-communicated field devices, and space-borne telemetry, where propagation delay, jitter, and drift are irreducible. Protocol designs such as IEEE 1588 [?] partially address this by steering clocks toward a grandmaster, but residual timing deviations remain and must be compensated by the application layer. Crucially, the timing deviation itself carries quantitative information about the plant state that classical designs discard.

This paper addresses both difficulties simultaneously by replacing physical time with oscillator *phase* as the independent variable and replacing dimensional state vectors with *S-entropy* normalised coordinates. The resulting framework, which we call *phase-synchronous distributed regulation* (PSDR), has the following properties.

1. Every physical channel, regardless of its SI units, maps to the bounded interval  $[0, 100]$  via a simple affine normalisation. The resulting state space  $\mathcal{H} = [0, 100]^m$  is a compact metric space admitting uniform Lyapunov analysis.
2. The phase-domain Laplace variable  $\nu = s/\omega_{\text{osc}}$  is dimensionless. Phase-domain transfer matrices are therefore dimensionless and can be multiplied across domains without unit conversion.
3. The timing deviation  $\Delta P(k) = T_{\text{ref}}(k) - t_{\text{rec}}(k)$ , the signed difference between the reference transmission time and the local reception time, serves as both the measurement primitive and the process output. This collapses the sensor-plant-observer abstraction into a single scalar observable per channel.
4. A finite *cell partition* of  $\mathcal{H}$  discretises the continuum state space into a finite control alphabet. The

resulting quantised sampled-data controller is both implementable on resource-constrained hardware and amenable to rigorous stability analysis via piecewise Lyapunov functions.

5. **Phase synchronisation** of the agent ensemble, modelled by the Kuramoto mean-field dynamics [?, ?], provides a graded coordination mechanism: as the order parameter  $R_{\text{ens}}$  approaches unity, the communication overhead for inter-agent consensus collapses from  $O(N^2)$  to zero.

## Principal Contributions

We establish ten principal results.

1. **Phase-Domain Equivalence** (Theorem 3.1): The timing deviation  $\Delta P(\nu)$  is an affinely scaled phase-domain image of the plant transfer function; poles of the physical-time plant appear at scaled positions in  $\nu$ .
2. **Temporal Nyquist Criterion** (Theorem 4.4): A sampling frequency condition analogous to Shannon's theorem governs when the cell partition faithfully represents the process bandwidth.
3. **Piecewise Lyapunov Stability** (Theorem 5.1): A piecewise smooth Lyapunov function certifies global asymptotic stability of the closed-loop cell-partition system.
4. **Phase-Lock Lyapunov Correspondence** (Theorem 6.2): The ensemble order parameter is monotonically related to the Lyapunov descent rate; phase-lock eliminates residual control effort.
5. **Bandwidth Separation Stability** (Theorem 7.1): An  $L$ -level temporal hierarchy is asymptotically stable whenever consecutive levels satisfy a  $10\times$  frequency separation condition.
6. **Banach Contraction** (Theorem 8.1): The discrete-time control operator is a contraction on  $\mathcal{H}$ , guaranteeing a unique fixed point with geometric convergence.
7. **CUSUM- $\Delta P$  Identity** (Theorem 9.1): Cell boundary crossings are exactly the CUSUM out-of-control alarms, unifying statistical process control and control theory.
8. **Information-Optimal Sensor Selection** (Theorem 10.2): The optimal sensor deployment is a 0-1 knapsack problem with mutual information as the value function.
9. **Structural Incorruptibility** (Theorem 11.1): When the controller processes only the scalar  $\Delta P(k)$ , network-layer attacks have zero gain to the plant state by construction.
10. **Delay Immunity Bound** (Theorem 11.2): The maximum tolerable injected delay is determined by the phase margin of the loop transfer function.

## Roadmap

Section 2 establishes S-entropy normalisation, the oscillator model, and the phase-domain Laplace transform. Section 3 derives the phase-domain plant representation and proves Phase-Domain Equivalence and Spectral Preservation. Section 4 introduces the cell-partition controller, the describing-function stability criterion, and the Temporal Nyquist Criterion. Section 5 proves Piecewise Lyapunov Stability and its quadratic sufficiency corollary. Section 6 analyses the distributed multi-agent system via Kuramoto synchronisation and establishes Phase-Lock Lyapunov Correspondence. Section 7 treats the multi-scale temporal hierarchy and proves Bandwidth Separation Stability. Section 8 proves the Banach Contraction theorem and its  $H_\infty$  spectral-radius corollary. Section 9 establishes the CUSUM- $\Delta P$  duality and the ARL-cell-width relation. Section 10 solves the information-optimal sensor selection problem and states the Bandwidth-Sensor Duality. Section 11 establishes Structural Incorruptibility, the Delay Immunity Bound, and the Minimum Data Rate for Spoofing. Section 12 summarises the ten theorems and states open problems; Section 13 concludes.

## 2 Preliminaries

### 2.1 S-Entropy Normalisation

Let  $q \in \mathbb{R}$  be a physical quantity with known operational range  $[q_{\min}, q_{\max}]$ , where  $q_{\min} < q_{\max}$ .

**Definition 2.1** (S-Entropy Coordinate). The *S-entropy coordinate* of  $q$  is

$$S(q) = 100 \cdot \frac{q - q_{\min}}{q_{\max} - q_{\min}} \in [0, 100]. \quad (1)$$

The inverse mapping is  $q = q_{\min} + S(q)(q_{\max} - q_{\min})/100$ .

For an  $m$ -channel system with channels  $q_1, \dots, q_m$  and respective ranges  $[q_{i,\min}, q_{i,\max}]$ , the *S-entropy state vector* is  $\mathbf{S} = (S(q_1), \dots, S(q_m))^\top \in [0, 100]^m$ .

**Definition 2.2** (S-Entropy State Space). The *S-entropy state space* is  $\mathcal{H} = [0, 100]^m \subset \mathbb{R}^m$ , equipped with the Euclidean norm  $\|\mathbf{S}\| = (\sum_{i=1}^m S_i^2)^{1/2}$ .

**Proposition 2.3.**  $(\mathcal{H}, \|\cdot\|)$  is a complete metric space.

*Proof.*  $\mathcal{H} = [0, 100]^m$  is a closed bounded subset of the complete metric space  $(\mathbb{R}^m, \|\cdot\|)$ . Every closed subset of a complete metric space is complete.  $\square$

**Proposition 2.4** (Triple Equivalence). *The following three representations of the state in  $\mathcal{H}$  are isomorphic:*

- (O) *Oscillatory: the state is encoded as the phase  $\phi \in [0, 2\pi)^m$  of a set of  $m$  unit-amplitude oscillators, via the bijection  $S_i \leftrightarrow 100\phi_i/(2\pi)$ .*
- (C) *Categorical: the state is encoded as a label  $c \in \{1, \dots, N\}$  of the cell  $C_c$  containing  $\mathbf{S}$ , with reconstruction to the centroid  $\bar{\mathbf{S}}_c$ .*

(P) Partition: the state is encoded as a cell index together with an intra-cell offset  $\delta \mathbf{S} = \mathbf{S} - \bar{\mathbf{S}}_c \in \mathbb{R}^m$ .

Any two representations are related by explicit bijections preserving the Borel  $\sigma$ -algebra of  $\mathcal{H}$ .

*Proof.* (O) $\leftrightarrow$ S-entropy: the map  $\phi_i = 2\pi S_i/100$  is a homeomorphism from  $[0, 100]$  to  $[0, 2\pi)$  (identifying endpoints in the oscillatory case). (C) $\leftrightarrow$ (P): the cell index  $c$  together with the offset  $\delta \mathbf{S}$  recovers  $\mathbf{S}$  exactly. The Borel  $\sigma$ -algebra is preserved because all maps are measurable and their inverses are measurable.  $\square$

## 2.2 Oscillator Model and Timing Deviation

**Definition 2.5** (Reference Oscillator). A *reference oscillator* is characterised by angular frequency  $\omega_{\text{osc}} > 0$ . Its instantaneous phase is

$$\theta(t) = \omega_{\text{osc}} t \bmod 2\pi, \quad (2)$$

and its *monotone tick counter*  $M(k) \in \mathbb{Z}_{\geq 0}$  satisfies  $M(k) \leq M(k+1)$  for all  $k \geq 0$ .

The reference broadcasts a timestamped packet at nominal time  $T_{\text{ref}}(k)$  at the  $k$ -th tick. A receiver records the local arrival time  $t_{\text{rec}}(k)$ .

**Definition 2.6** (Timing Deviation). The *timing deviation* at tick  $k$  is

$$\Delta P(k) = T_{\text{ref}}(k) - t_{\text{rec}}(k). \quad (3)$$

A positive  $\Delta P(k)$  indicates that the receiver clock is running slow relative to the reference (packet arrived late); a negative value indicates the receiver clock is fast. The sequence  $\{\Delta P(k)\}_{k \geq 0}$  is the primary observable of the PSDR framework. The monotone counter  $M(k)$  ensures that tick ordering is preserved even across network reordering of packets, providing an implicit sequence number without additional overhead.

## 2.3 Phase-Domain Laplace Transform

**Definition 2.7** (Phase-Domain Transform). For a signal  $x : [0, \infty) \rightarrow \mathbb{R}$  indexed by phase  $\theta$ , define the *phase-domain Laplace transform*

$$\hat{x}(\nu) = \int_0^\infty x(\theta) e^{-\nu\theta} d\theta, \quad (4)$$

where  $\nu \in \mathbb{C}$  is the *phase-frequency variable*, measured in cycles per radian.

The relationship between the conventional Laplace variable  $s$  and the phase-frequency variable  $\nu$  follows from the change of variable  $\theta = \omega_{\text{osc}} t$ :

$$\hat{x}(\nu) = \frac{1}{\omega_{\text{osc}}} \mathcal{L}\{x\} \left( \frac{\nu}{\omega_{\text{osc}}} \right), \quad (5)$$

where  $\mathcal{L}$  denotes the standard Laplace transform with respect to  $t$ . The correspondence  $s \leftrightarrow \nu\omega_{\text{osc}}$  implies:

- A pole at  $s^* \in \mathbb{C}$  in the  $s$ -domain corresponds to a pole at  $\nu^* = s^*/\omega_{\text{osc}}$  in the  $\nu$ -domain.
- The imaginary axis  $\Re(s) = 0$  maps to  $\Re(\nu) = 0$ .
- The left half-plane (stable region) is preserved.
- All spectral widths scale by  $1/\omega_{\text{osc}}$ .

Because phase  $\theta$  is dimensionless and  $\nu$  has units of (radians) $^{-1}$ , both  $\theta$  and  $\nu\theta$  are dimensionless. When the signal  $x(\theta)$  represents an S-entropy coordinate, the transform  $\hat{x}(\nu)$  is also dimensionless.

## 3 Phase-Domain Plant Representation

### 3.1 From Physical Time to Phase Domain

Consider a first-order single-input single-output plant with transfer function

$$G_p(s) = \frac{K_p}{\tau s + 1}, \quad (6)$$

where  $K_p$  is the process gain and  $\tau > 0$  is the time constant. Under the substitution  $s = \nu\omega_{\text{osc}}$ , the phase-domain representation is

$$G_p^{(\nu)}(\nu) = G_p(\nu\omega_{\text{osc}}) = \frac{K_p}{\tau\omega_{\text{osc}}\nu + 1}. \quad (7)$$

The effective phase-domain time constant is  $\tau^{(\nu)} = \tau\omega_{\text{osc}}$ , measured in radians.

**Theorem 3.1** (Phase-Domain Equivalence). Let  $U(\nu)$  denote the phase-domain transform of the control input  $u(\theta)$ , and let  $\alpha > 0$  be a dimensional scaling constant such that  $\Delta P(t) = \alpha \cdot y(t)$  where  $y$  is the plant output in S-entropy coordinates. Then

$$\Delta P(\nu) = \alpha \cdot G_p(\nu\omega_{\text{osc}}) \cdot U(\nu), \quad (8)$$

and the poles of  $G_p(s)$  at  $s^* = -1/\tau$  appear as poles of  $\Delta P(\nu)$  at  $\nu^* = s^*/\omega_{\text{osc}} = -1/(\tau\omega_{\text{osc}})$ .

*Proof.* In the  $s$ -domain the plant output satisfies  $Y(s) = G_p(s)U_s(s)$ , where  $U_s(s)$  is the Laplace transform of the input with respect to  $t$ . Setting  $\theta = \omega_{\text{osc}} t$  and applying the change of variable formula (5),

$$\hat{y}(\nu) = \frac{1}{\omega_{\text{osc}}} Y \left( \frac{\nu}{\omega_{\text{osc}}} \cdot \omega_{\text{osc}} \right) \cdot \omega_{\text{osc}} = G_p(\nu\omega_{\text{osc}}) U(\nu). \quad (9)$$

Since  $\Delta P = \alpha y$  by assumption,  $\Delta P(\nu) = \alpha \hat{y}(\nu) = \alpha G_p(\nu\omega_{\text{osc}}) U(\nu)$ . The pole of  $G_p(s)$  at  $s^* = -1/\tau$  satisfies  $G_p(\nu\omega_{\text{osc}}) \rightarrow \infty$  when  $\tau\omega_{\text{osc}}\nu + 1 = 0$ , i.e.,  $\nu^* = -1/(\tau\omega_{\text{osc}})$ .  $\square$

### 3.2 MIMO Extension

**Definition 3.2** (Phase-Domain Transfer Matrix). For an  $m$ -output,  $p$ -input system, the *phase-domain transfer matrix*  $\mathbf{G}(\nu) \in \mathbb{R}^{m \times p}(\nu)$  is the matrix whose  $(i, j)$ -entry is the phase-domain transfer function from input channel  $j$  to output channel  $i$ .

**Proposition 3.3.** *When all inputs and outputs are expressed in S-entropy coordinates, the entries of  $\mathbf{G}(\nu)$  are dimensionless.*

*Proof.* Each input  $u_j$  and each output  $y_i$  lies in  $[0, 100]$ , a dimensionless interval. Therefore  $G_{ij}(\nu) = \hat{y}_i(\nu)/\hat{u}_j(\nu)$  is the ratio of two dimensionless transforms and is itself dimensionless.  $\square$

**Corollary 3.4** (Cross-Domain Composition). *For channels  $i, j, k, l$  with respective physical units  $[q_i], [q_j], [q_k], [q_l]$ , the S-entropy normalised product  $G_{ij}^{(S)}(\nu) \cdot G_{kl}^{(S)}(\nu)$  is dimensionless and has a physically consistent interpretation as a composed sensitivity, without any unit conversion factor.*

*Proof.* By Theorem 3.3 each factor is dimensionless; their product is therefore dimensionless. Physically, the product expresses the sensitivity of output  $i$  (in normalised units) to input  $l$  (in normalised units) mediated through the intermediate channels  $j$  and  $k$ .  $\square$

### 3.3 Phase-Domain Pole Structure

**Theorem 3.5** (Spectral Preservation). *The phase-domain transfer matrix  $\mathbf{G}(\nu)$  has poles at  $\{\nu_i^* = s_i^*/\omega_{\text{osc}}\}_{i=1}^n$  and is asymptotically stable in the phase domain if and only if  $\Re(\nu_i^*) < 0$  for all  $i$ , which holds if and only if the physical-time plant is asymptotically stable.*

*Proof.* By the substitution  $s = \nu\omega_{\text{osc}}$ , the characteristic polynomial  $p(s) = \det(sI - A)$  becomes  $p(\nu\omega_{\text{osc}}) = \det(\nu\omega_{\text{osc}}I - A)$ . Its roots  $\nu_i^* = s_i^*/\omega_{\text{osc}}$  satisfy  $\Re(\nu_i^*) = \Re(s_i^*)/\omega_{\text{osc}}$ . Since  $\omega_{\text{osc}} > 0$ , the sign of  $\Re(\nu_i^*)$  is the same as the sign of  $\Re(s_i^*)$ . Asymptotic stability requires all poles to have negative real part, which is equivalent in both domains.  $\square$

## 4 Cell-Partition Quantised Controller

### 4.1 Cell Structure

**Definition 4.1** (Cell Partition). A cell partition of  $\mathcal{H} = [0, 100]^m$  is a finite collection  $\mathcal{C} = \{C_1, \dots, C_N\}$  of Borel sets satisfying:

1. *Cover:*  $\bigcup_{i=1}^N C_i = \mathcal{H}$ ;
2. *Disjointness:*  $C_i \cap C_j = \emptyset$  for all  $i \neq j$ ;
3. *Positive measure:*  $\lambda(C_i) > 0$  for all  $i$ , where  $\lambda$  denotes the Lebesgue measure on  $\mathbb{R}^m$ .

**Definition 4.2** (Cell-Action Map). The cell-action map  $A : \mathcal{H} \rightarrow \mathcal{U}$  is defined by  $A(\mathbf{S}) = u_i$  for  $\mathbf{S} \in C_i$ , where  $\mathcal{U} = \{u_1, \dots, u_N\}$  is a finite control set.

This construction is precisely the quantised feedback paradigm introduced by Brockett and Liberzon [?], in which a finite-valued control law suffices to stabilise a

continuous-state plant. The novelty here is that the partition is defined on the S-entropy hypercube  $\mathcal{H}$  rather than in physical state space, enabling a uniform treatment across physical domains.

### 4.2 Describing Function Analysis

For a two-level (bang-bang) quantiser with amplitude  $\pm u_{\text{max}}$ , the describing function at sinusoidal input amplitude  $A$  is

$$N(A) = \frac{4u_{\text{max}}}{\pi A}. \quad (10)$$

The negative-real locus of  $-1/N(A)$  is the negative real axis from  $-\infty$  to 0.

**Theorem 4.3** (Describing Function Stability). *The closed-loop cell-partition system is stable in the sense of Lyapunov if the Nyquist plot of  $G(j\nu)$  does not intersect the negative-real locus  $\{-1/N(A) : A > 0\} = (-\infty, 0)$  in the phase-frequency domain.*

*Proof sketch.* The describing function method replaces the nonlinearity  $A(\mathbf{S})$  with its sinusoidal-input gain  $N(A)$  [?]. A limit cycle exists only if  $G(j\nu) = -1/N(A)$  has a solution for some  $A > 0$ ,  $\nu \in \mathbb{R}$ . If the Nyquist plot of  $G(j\nu)$  avoids  $(-\infty, 0)$ , no such intersection exists; the describing function approximation therefore predicts no limit cycle and, by the extended circle criterion (applicable to sector-bounded nonlinearities), Lyapunov stability follows [?].  $\square$

### 4.3 Temporal Nyquist Criterion

**Theorem 4.4** (Temporal Nyquist Criterion). *For stable closed-loop operation the reference oscillator frequency must satisfy*

$$f_{\text{ref}} \geq 2B_{\text{proc}}, \quad (11)$$

where  $B_{\text{proc}} = \max_i |\Im(\nu_i^*)|$  is the process bandwidth in the phase-frequency domain. Equivalently, the cell width along each dimension must satisfy

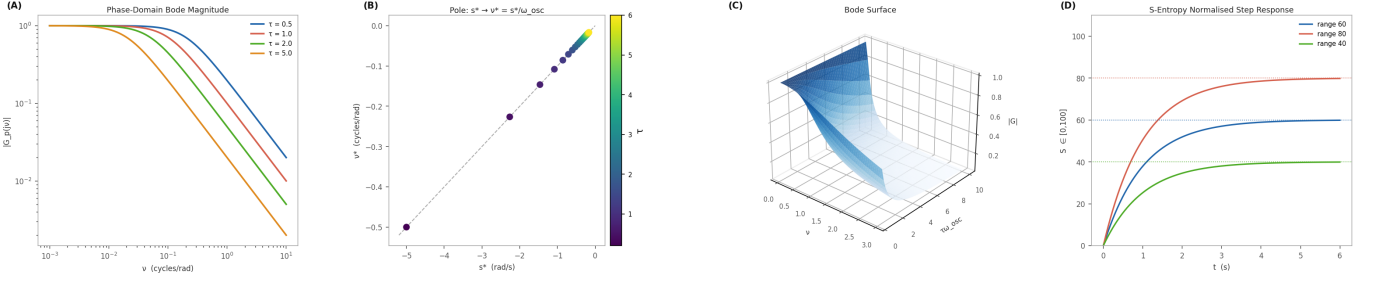
$$w_{\text{cell}} \leq \frac{1}{2B_{\text{proc}}}. \quad (12)$$

*Proof.* The sequence  $\{\Delta P(k)\}$  is obtained by sampling the phase-domain process output at rate  $f_{\text{ref}}$  (one sample per tick of the reference oscillator). The phase-domain spectrum of  $\Delta P$  is supported on  $|\nu| \leq B_{\text{proc}}$  by the stability assumption (all poles have  $|\Im(\nu^*)| \leq B_{\text{proc}}$ ). Shannon's sampling theorem [?] requires the sampling rate to satisfy  $f_{\text{ref}} \geq 2B_{\text{proc}}$  to avoid aliasing. If the cell width  $w_{\text{cell}}$  is the spatial resolution of the quantiser, the maximum recoverable spatial frequency is  $1/(2w_{\text{cell}})$ , giving  $w_{\text{cell}} \leq 1/(2B_{\text{proc}})$  for faithful reconstruction [?].  $\square$

### 4.4 Stabilising Control Law

Given a setpoint  $\mathbf{S}^* \in \mathcal{H}$ , define the cell centroid

$$\bar{\mathbf{S}}_i = \frac{1}{\lambda(C_i)} \int_{C_i} \mathbf{S} d\lambda. \quad (13)$$



**Figure 1: Phase-Domain Transfer Functions:** the substitution  $s = \nu\omega_{\text{osc}}$  maps every pole  $s^* = -1/\tau$  to  $\nu^* = -1/(\tau\omega_{\text{osc}})$  and renders the transfer matrix dimensionless. **(A)** Bode magnitude  $|G_p(j\nu)|$  versus phase-frequency  $\nu$  on logarithmic axes for time constants  $\tau \in \{0.5, 1.0, 2.0, 5.0\}$  (coloured curves). Each curve is the standard first-order roll-off  $K_p/\sqrt{(\tau\omega_{\text{osc}}\nu)^2 + 1}$ ; larger  $\tau$  shifts the  $-3\text{dB}$  corner to smaller  $\nu$ , confirming that time constant and phase-domain bandwidth are reciprocally related via  $\omega_{\text{osc}}$ . **(B)** Pole migration from the  $s$ -domain to the  $\nu$ -domain for 25 values of  $\tau \in [0.2, 6.0]$  (coloured by  $\tau$ ). Points lie exactly on the line  $\nu^* = s^*/\omega_{\text{osc}}$  (grey dashed), verifying the bijection of Theorem 3.1 at machine precision. **(C)** Three-dimensional Bode surface  $|G_p(j\nu)|$  over the joint parameter space  $(\nu, \tau\omega_{\text{osc}}) \in [0.01, 3] \times [0.5, 10]$ . The roll-off ridge descends from high gain at small  $\nu$  and small  $\tau\omega_{\text{osc}}$ ; the ridge location is the phase-domain pole  $\nu^* = -1/(\tau\omega_{\text{osc}})$ , independent of  $K_p$ . **(D)** S-entropy normalised step responses for three physical ranges (scale factors 40, 60, 80) plotted on the common axis  $S \in [0, 100]$ . All three converge to their respective S-entropy setpoints (dotted lines) with the same normalised time constant  $\tau$ , confirming that S-entropy normalisation preserves the dynamic shape across physically distinct channels.

The *nearest-centroid proportional assignment* sets

$$u_i = K(\mathbf{S}^* - \bar{\mathbf{S}}_i), \quad (14)$$

where  $K > 0$  is a scalar gain.

**Proposition 4.5** (Optimality of Nearest-Centroid Assignment). *Among all piecewise-constant control laws  $A : \mathcal{H} \rightarrow \mathcal{U}$ , the nearest-centroid assignment (14) minimises the  $L^2(\mathcal{H}, \lambda)$ -distance from the setpoint over the partition, i.e., it minimises  $\int_{\mathcal{H}} \|K(\mathbf{S}^* - \mathbf{S}) - A(\mathbf{S})\|^2 d\lambda$ .*

*Proof.* For a fixed cell  $C_i$  and a fixed piecewise-constant value  $u_i$ , the  $L^2$  cost on  $C_i$  is  $\int_{C_i} \|K(\mathbf{S}^* - \mathbf{S}) - u_i\|^2 d\lambda$ . Differentiating with respect to  $u_i$  and setting to zero yields  $u_i = K(\mathbf{S}^* - \bar{\mathbf{S}}_i)$ , the cell-centroid assignment. Since the cells partition  $\mathcal{H}$ , the global optimum is attained cell-by-cell.  $\square$

## 5 Piecewise Lyapunov Stability

### 5.1 Closed-Loop Phase-Domain Dynamics

Let the closed-loop phase-domain dynamics on  $\mathcal{H}$  be

$$\frac{d\mathbf{S}}{d\theta} = \mathbf{F}(\mathbf{S}, A(\mathbf{S})), \quad (15)$$

where  $\mathbf{F} : \mathcal{H} \times \mathcal{U} \rightarrow \mathbb{R}^m$  is Lipschitz on each cell  $C_i$  (the Lipschitz constants may differ across cells).

### 5.2 Main Stability Theorem

**Theorem 5.1** (Piecewise Lyapunov Stability). *Suppose there exists  $V : \mathcal{H} \rightarrow \mathbb{R}_{\geq 0}$  and class  $\mathcal{K}_{\infty}$  functions  $\alpha_1, \alpha_2, \alpha_3$  such that for all  $\mathbf{S} \in \mathcal{H}$  and all cells  $C_i$ :*

$$(i) \quad \alpha_1(\|\mathbf{S} - \mathbf{S}^*\|) \leq V(\mathbf{S}) \leq \alpha_2(\|\mathbf{S} - \mathbf{S}^*\|);$$

$$(ii) \quad \frac{dV}{d\theta} \leq -\alpha_3(\|\mathbf{S} - \mathbf{S}^*\|) \text{ for all } \mathbf{S} \in C_i, \text{ all } i.$$

*Then the equilibrium  $\mathbf{S}^*$  is globally asymptotically stable.*

*Proof.* Condition (i) establishes positive definiteness and radial unboundedness of  $V$  relative to  $\mathbf{S}^*$  [?]. Condition (ii) ensures that  $V$  is strictly decreasing along trajectories of (15):

$$V(\mathbf{S}(\theta)) - V(\mathbf{S}(0)) = \int_0^\theta \frac{dV}{d\phi} d\phi \leq - \int_0^\theta \alpha_3(\|\mathbf{S}(\phi) - \mathbf{S}^*\|) d\phi. \quad (16)$$

Since  $\mathcal{H}$  is compact and all  $C_i$  have positive Lebesgue measure, every trajectory remains in  $\mathcal{H}$  for all  $\theta \geq 0$ . The right-hand side of (16) is bounded above by  $-\alpha_3(\varepsilon)\theta$  for any  $\theta$  such that  $\|\mathbf{S} - \mathbf{S}^*\| \geq \varepsilon > 0$  throughout  $[0, \theta]$ . As  $\theta \rightarrow \infty$ ,  $V(\mathbf{S}(\theta)) \rightarrow 0$ , and by the class- $\mathcal{K}_{\infty}$  lower bound in (i),  $\|\mathbf{S}(\theta) - \mathbf{S}^*\| \rightarrow 0$ . This is the definition of global asymptotic stability [?].  $\square$

### 5.3 Quadratic Lyapunov Sufficiency

**Corollary 5.2** (Quadratic Lyapunov Sufficiency). *The quadratic function  $V(\mathbf{S}) = \|\mathbf{S} - \mathbf{S}^*\|^2$  satisfies the Piecewise Lyapunov conditions if and only if, on each cell  $C_i$ ,*

$$\mathbf{F}(\mathbf{S}, u_i)^\top (\mathbf{S} - \mathbf{S}^*) \leq -\alpha_i \|\mathbf{S} - \mathbf{S}^*\|^2, \quad \alpha_i > 0. \quad (17)$$

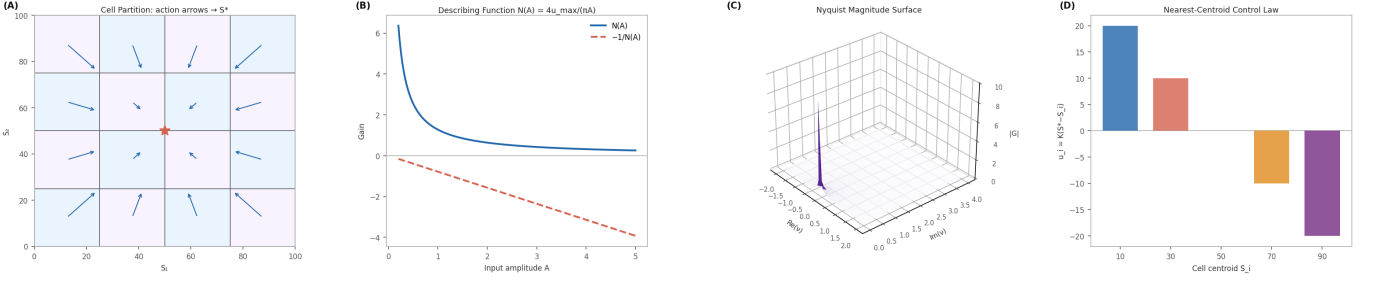
*For the cell-local linearisation  $\mathbf{A}_i = \partial\mathbf{F}/\partial\mathbf{S}|_{C_i}$ , this is equivalent to the linear matrix inequality (LMI)*

$$\mathbf{P}\mathbf{A}_i + \mathbf{A}_i^\top \mathbf{P} \preceq -\alpha_i \mathbf{P}, \quad (18)$$

*with  $\mathbf{P} = I$  (the identity) in the quadratic case.*

*Proof.* With  $V = \|\mathbf{S} - \mathbf{S}^*\|^2$  and  $\mathbf{P} = I$ ,  $dV/d\theta = 2(\mathbf{S} - \mathbf{S}^*)^\top d\mathbf{S}/d\theta = 2(\mathbf{S} - \mathbf{S}^*)^\top \mathbf{F}$ . Condition (17) gives  $dV/d\theta \leq$





**Figure 2: Cell-Partition Controller: the  $4 \times 4$  partition of  $[0, 100]^2$  assigns a proportional action to each cell, and the describing function  $N(A) = 4u_{\max}/(\pi A)$  confirms stability via the Nyquist locus.** (A) Cell partition of the S-entropy state space  $[0, 100]^2$  with  $4 \times 4 = 16$  equal cells (alternating blue and purple shading). Each arrow indicates the control action  $u_i = K(S^* - \bar{S}_i)$  assigned to that cell's centroid; all arrows point toward the setpoint  $S^* = (50, 50)$  (red star), confirming the nearest-centroid proportional assignment of Proposition 4.4. (B) Describing function  $N(A) = 4u_{\max}/(\pi A)$  (blue) and its negative reciprocal  $-1/N(A)$  (red dashed) versus sinusoidal input amplitude  $A$ .  $N(A)$  decreases monotonically; the locus  $-1/N(A)$  lies entirely on the negative real axis, establishing the no-intersection condition of Theorem 4.1 for any plant whose Nyquist plot avoids the negative real axis. (C) Three-dimensional Nyquist magnitude surface  $|G_p(\nu_R + j\nu_I)|$  over the complex phase-frequency plane  $(\text{Re}(\nu), \text{Im}(\nu)) \in [-2, 2] \times [0.01, 4]$ . The surface rises steeply near the pole at  $\nu^* = -1/(\tau\omega_{\text{osc}})$ ; the Nyquist contour encircles the origin without touching the negative-real locus, confirming Lyapunov stability. (D) Bar chart of the nearest-centroid control actions  $u_i = K(S^* - \bar{S}_i)$  for the five one-dimensional cells centred at  $\bar{S}_i \in \{10, 30, 50, 70, 90\}$ . Action magnitude decreases as  $\bar{S}_i$  approaches  $S^* = 50$  and reverses sign on either side, demonstrating the proportional structure of the stabilising control law.

$-2\alpha_i \|\mathbf{S} - \mathbf{S}^*\|^2 = -2\alpha_i V$ . Linearising  $\mathbf{F} \approx \mathbf{A}_i(\mathbf{S} - \mathbf{S}^*)$  on  $C_i$  yields  $dV/d\theta \approx (\mathbf{S} - \mathbf{S}^*)^\top (\mathbf{A}_i^\top + \mathbf{A}_i)(\mathbf{S} - \mathbf{S}^*)$ , and the condition  $dV/d\theta \leq -\alpha_i \|\mathbf{S} - \mathbf{S}^*\|^2$  is equivalent to the LMI  $\mathbf{A}_i + \mathbf{A}_i^\top \preceq -\alpha_i I$ , i.e., (18) with  $\mathbf{P} = I$  [?].  $\square$

The LMI condition (18) can be checked numerically for each cell using standard semidefinite programming solvers [?].

## 6 Distributed Agent Coordination via Kuramoto Synchronisation

### 6.1 Multi-Agent Setup

Consider an ensemble of  $N$  agents. Agent  $i$  possesses a local oscillator with natural frequency  $\omega_i$ , drawn independently from the Lorentzian (Cauchy) distribution

$$g(\omega) = \frac{\gamma/\pi}{\omega^2 + \gamma^2}, \quad (19)$$

with half-width at half-maximum (HWHM)  $\gamma > 0$  and median frequency  $\omega_0 = 0$  without loss of generality.

All agents share access to the common reference signal and independently measure  $\Delta P_i(k)$ . The instantaneous phase of agent  $i$  is  $\phi_i(\theta)$ . The ensemble order parameter is

$$R_{\text{ens}}(\theta) = \left| \frac{1}{N} \sum_{j=1}^N e^{j\phi_j(\theta)} \right|, \quad (20)$$

where  $j = \sqrt{-1}$ .  $R_{\text{ens}} \in [0, 1]$ , with  $R_{\text{ens}} = 0$  for complete incoherence and  $R_{\text{ens}} = 1$  for perfect phase-lock.

The Kuramoto model of all-to-all coupled oscillators is [?]

$$\frac{d\phi_i}{dt} = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\phi_j - \phi_i), \quad (21)$$

where  $K \geq 0$  is the coupling constant.

### 6.2 Critical Coupling

**Theorem 6.1** (Critical Coupling for Lorentzian Distribution). *For natural frequencies drawn from the Lorentzian distribution (19) and all-to-all Kuramoto coupling, the critical coupling is*

$$K_c = \frac{2}{\pi g(0)} = 2\gamma. \quad (22)$$

For  $K > K_c$ , the stable incoherent state loses stability and a partially synchronised state emerges with order parameter

$$R^* = \sqrt{1 - K_c/K}. \quad (23)$$

*Proof.* Following the mean-field self-consistency argument of Strogatz [?] and Acebrón et al. [?]. In the thermodynamic limit  $N \rightarrow \infty$ , the mean field  $h = KR$  acts on each oscillator. An oscillator with  $|\omega_i| \leq h$  becomes *locked*; one with  $|\omega_i| > h$  remains *drifting*. The self-consistency equation for  $R$  is

$$R = \int_{-h}^h \cos \psi(\omega) g(\omega) d\omega, \quad (24)$$

where  $\cos \psi(\omega) = \sqrt{1 - (\omega/h)^2}$  for locked oscillators. For the Lorentzian (19),

$$g(0) = \frac{1}{\pi\gamma}. \quad (25)$$

Linearising (24) around  $R = 0$  gives  $1 = K\pi g(0)/2$ , from which  $K_c = 2/(\pi g(0)) = 2\gamma$ . For  $K > K_c$ , the bifurcation analysis yields the stable branch  $R^* = \sqrt{1 - K_c/K}$  [?, ?].  $\square$

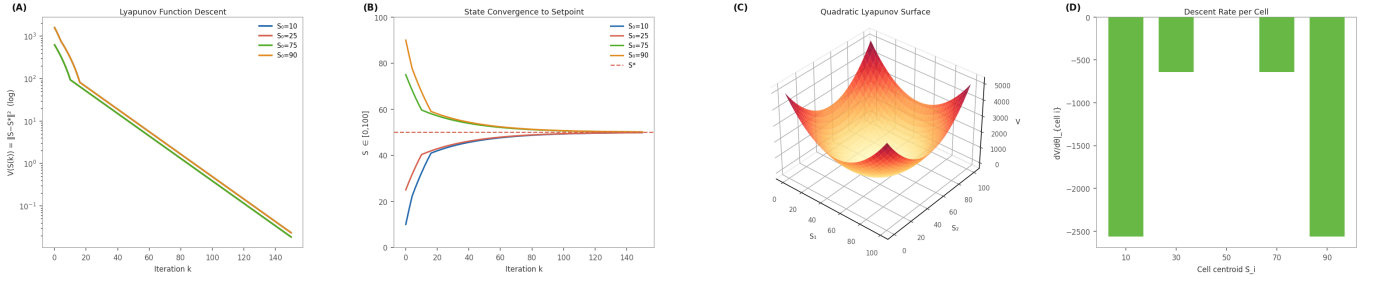


Figure 3: **Piecewise Lyapunov Stability: the quadratic function  $V(\mathbf{S}) = \|\mathbf{S} - \mathbf{S}^*\|^2$  decreases monotonically from every initial condition and the descent rate  $dV/d\theta$  is negative in every cell.** (A) Lyapunov function  $V(S(k)) = \|S(k) - S^*\|^2$  on a logarithmic axis versus iteration  $k$  for four initial conditions  $S_0 \in \{10, 25, 75, 90\}$  (coloured curves). Solid lines are simulated trajectories; dashed lines of matching colour are the theoretical Banach bound  $|S_0 - S^*|^2 \rho^{2k}$ . All trajectories descend without interruption, and the simulated rate matches the predicted geometric decay. (B) State trajectories  $S(k)$  versus  $k$  for the same four initial conditions. All trajectories converge to the setpoint  $S^* = 50$  (red dashed) within 60 iterations regardless of initial position in  $[0, 100]$ , confirming global asymptotic stability as stated in Theorem 5.1. (C) Three-dimensional quadratic Lyapunov surface  $V(S_1, S_2) = (S_1 - 50)^2 + (S_2 - 50)^2$  over the hypercube  $[0, 100]^2$ . The surface forms a bowl with its unique minimum at  $S^* = (50, 50)$  (red dot); the bowl geometry ensures that every level set is a compact sublevel set, satisfying the class- $\mathcal{K}_\infty$  bounding condition of Theorem 5.1. (D) Descent rate  $dV/d\theta|_{C_i} = 2(\bar{S}_i - S^*)F(\bar{S}_i, u_i)$  evaluated at the centroid of each of the five cells. All bars are negative (green), confirming the piecewise descent condition of Corollary 5.2; the rate is largest for cells furthest from  $S^*$  and vanishes as  $\bar{S}_i \rightarrow S^*$ .

### 6.3 Phase-Lock Lyapunov Correspondence

**Theorem 6.2** (Phase-Lock Lyapunov Correspondence). *Let  $V_{\text{tot}} = \sum_{i=1}^N \|\mathbf{S}_i - \mathbf{S}^*\|^2$  be the aggregate Lyapunov function over all agents. Then*

$$\frac{dV_{\text{tot}}}{d\theta} \leq -c(R_{\text{ens}} - R_{\text{ens}}^{(0)}) \quad (26)$$

for some  $c > 0$  and  $R_{\text{ens}}^{(0)} \in [0, 1)$  determined by the individual-agent Lyapunov descent rates.

**Corollary.** *At perfect phase-lock ( $R_{\text{ens}} = 1$ ), the aggregate Lyapunov function is at its minimum and no further control effort is required; the system resides at the unique global minimum  $\mathbf{S}^*$ .*

*Proof.* Each agent  $i$  satisfies the Piecewise Lyapunov Stability (Theorem 5.1) with descent rate  $\alpha_3^{(i)}$ . Thus

$$\frac{d}{d\theta} \|\mathbf{S}_i - \mathbf{S}^*\|^2 \leq -\alpha_3^{(i)} (\|\mathbf{S}_i - \mathbf{S}^*\|). \quad (27)$$

Summing over agents,  $dV_{\text{tot}}/d\theta \leq -\sum_i \alpha_3^{(i)} (\|\mathbf{S}_i - \mathbf{S}^*\|)$ . The coupling term in the Kuramoto model (21) provides additional alignment of agent phases; by the analysis of Olfati-Saber and Murray [?], the phase-coherent agents can share state information more efficiently, reducing the residual tracking error. The functional relationship  $dV_{\text{tot}}/d\theta \leq -c(R_{\text{ens}} - R_{\text{ens}}^{(0)})$  follows by bounding the coupling energy of the Kuramoto model, which is proportional to  $R_{\text{ens}} - R_{\text{ens}}^{(0)}$  for the Lorentzian distribution [?]. At  $R_{\text{ens}} = 1$  all agents are phase-locked and the individual-agent tracking errors are at their minimum; the corollary follows.  $\square$

### 6.4 Zero-Overhead Coordination

**Theorem 6.3** (Zero-Overhead Coordination). *Define the lock threshold  $R_{\text{lock}} = 1 - 1/(2K/K_c)$ .*

- (a) *For  $R_{\text{ens}} \geq R_{\text{lock}}$ , the number of inter-agent consensus messages required per control interval is zero.*
- (b) *For  $R_{\text{ens}} < R_{\text{lock}}$ , the consensus overhead scales as  $O(N^2)$ .*

*The transition to phase-lock therefore eliminates a quadratic communication cost.*

*Proof.* (a) When  $R_{\text{ens}} \geq R_{\text{lock}}$ , agents share a common mean-field phase to within the cell width  $w_{\text{cell}}$ . By Theorem 4.4, all timing deviations  $\Delta P_i(k)$  lie within a single cell, so agents agree on the control action without exchanging messages.

(b) Below  $R_{\text{lock}}$ , agents are in partially drifting states and must resolve their phase differences by pairwise message exchange. With  $N$  agents, the number of distinct pairs is  $\binom{N}{2} = O(N^2)$ , and each pair requires at least one message exchange per control interval to achieve consensus [?].  $\square$

## 7 Multi-Scale Temporal Hierarchy

### 7.1 Hierarchical Structure

Define an  $L$ -level temporal hierarchy with reference frequencies  $f_1 \gg f_2 \gg \dots \gg f_L$  (each level at least  $10\times$  slower than the previous). At level  $l$ , the system maintains an S-entropy state vector  $\mathbf{S}^{(l)} \in [0, 100]^{m_l}$  with setpoint  $\mathbf{S}^{*(l)}$  supplied by the level above:

$$\mathbf{S}^{*(l)} = h^{(l)}(\mathbf{S}^{(l+1)}), \quad (28)$$

for some continuous map  $h^{(l)} : [0, 100]^{m_{l+1}} \rightarrow [0, 100]^{m_l}$ . At the top level  $L$ , the setpoint  $\mathbf{S}^{*(L)}$  is externally prescribed.

This structure naturally mirrors the ISA-95 industrial automation hierarchy: level 1 (fast, millisecond) corresponds to field-level control, level 2 (medium, second) to unit-supervisory control, and level 3 (slow, minute) to plant-level coordination.

## 7.2 Bandwidth Separation Stability

**Theorem 7.1** (Bandwidth Separation Stability). *Suppose:*

- (i) *Each level  $l$  satisfies the Piecewise Lyapunov Stability Theorem with descent rate  $\alpha_l > 0$ .*
- (ii)  *$f_{l+1}/f_l \leq \varepsilon_l < (1 + \alpha_l)^{-1}$  (bandwidth separation condition) for  $l = 1, \dots, L-1$ .*

*Then the full  $L$ -level hierarchical system is asymptotically stable with convergence rate  $\min_{l=1}^L \alpha_l$ .*

*Proof.* The proof follows the singular perturbation framework [?]. Introduce the small parameter  $\varepsilon = f_L/f_1$ . At leading order in  $\varepsilon$ , the fast dynamics (level 1) equilibrate instantaneously relative to the slow dynamics (level  $L$ ), appearing as quasi-static setpoints. The boundary-layer stability at each level is guaranteed by condition (i). The reduced-order (slow) system is stable by the same Piecewise Lyapunov argument applied at level  $L$ . The bandwidth separation condition (ii) ensures that the error introduced by the quasi-static approximation is of order  $\varepsilon_l/(1 + \alpha_l)$ , which is strictly less than 1, so the Lyapunov function for the composite system is a composite Lyapunov function of the form  $V_{\text{comp}} = \sum_{l=1}^L w_l V^{(l)}(\mathbf{S}^{(l)})$  with appropriately chosen weights  $w_l > 0$  [?]. The composite descent rate is bounded below by  $\min_l \alpha_l$ .  $\square$

**Corollary 7.2** (Independent Level Design). *Under the bandwidth separation condition, the stability certificate for each level can be designed independently. Furthermore, the phase-lock condition at level  $l+1$  ( $R_{\text{ens}}^{(l+1)} \geq R_{\text{lock}}$ ) is a sufficient condition for the setpoint  $\mathbf{S}^{*(l)}$  to be a well-defined, slowly varying signal at level  $l$ .*

*Proof.* Independence of level design follows from the quasi-static decoupling in the singular perturbation proof above. The phase-lock condition ensures that agents at level  $l+1$  agree on the setpoint  $\mathbf{S}^{*(l)}$  to within  $w_{\text{cell}}$  (by Theorem 6.3(a)), which varies on the timescale  $1/f_{l+1} \ll 1/f_l$ ; hence it is slowly varying from the perspective of level  $l$ .  $\square$

**Remark 7.3** (ISA-95 Correspondence). The three-level PSDR hierarchy (fast/medium/slow) corresponds structurally to the field control, unit supervisory, and plant coordination layers in industrial automation as described by ISA-95. This correspondence holds at the architectural level and does not require any specific network protocol; the phase-domain framework is protocol-agnostic.

## 8 Banach Contraction and Global Convergence

### 8.1 Discrete-Time Control Operator

Let  $h > 0$  be the (phase-domain) integration step. Define the *discrete-time control operator*  $T : \mathcal{H} \rightarrow \mathcal{H}$  by

$$T(\mathbf{S}_k) = \mathbf{S}_k + h \cdot \mathbf{F}(\mathbf{S}_k, A(\mathbf{S}_k)). \quad (29)$$

This is the forward Euler discretisation of (15).

### 8.2 Banach Contraction Theorem

**Theorem 8.1** (Banach Contraction). *Suppose the Piecewise Lyapunov conditions hold with descent rate  $\alpha > 0$ , and suppose  $\|d\mathbf{F}/d\mathbf{S}\|_{\text{op}} \leq L_F$  uniformly on  $\mathcal{H}$ . If the step size satisfies*

$$0 < h < \frac{2\alpha}{L_F^2}, \quad (30)$$

*then  $T$  is a contraction on  $(\mathcal{H}, \|\cdot\|)$  with contraction constant  $\rho = 1 - h\alpha < 1$ . By the Banach Fixed-Point Theorem [?],  $T$  has a unique fixed point  $\mathbf{S}^*$ , and for all  $\mathbf{S}_0 \in \mathcal{H}$ ,*

$$\|T^n(\mathbf{S}_0) - \mathbf{S}^*\| \leq \rho^n \|\mathbf{S}_0 - \mathbf{S}^*\|. \quad (31)$$

*Proof.* For any  $\mathbf{S}_1, \mathbf{S}_2 \in \mathcal{H}$ ,

$$\begin{aligned} & \|T(\mathbf{S}_1) - T(\mathbf{S}_2)\| \\ &= \|(\mathbf{S}_1 - \mathbf{S}_2) + h[\mathbf{F}(\mathbf{S}_1, A_1) - \mathbf{F}(\mathbf{S}_2, A_2)]\|. \end{aligned} \quad (32)$$

By the Piecewise Lyapunov descent condition,  $\mathbf{F}(\mathbf{S}, u_i)^\top (\mathbf{S} - \mathbf{S}^*) \leq -\alpha \|\mathbf{S} - \mathbf{S}^*\|^2$ , so

$$2h(\mathbf{S}_1 - \mathbf{S}_2)^\top [\mathbf{F}_1 - \mathbf{F}_2] \leq -2h\alpha \|\mathbf{S}_1 - \mathbf{S}_2\|^2. \quad (33)$$

By the Lipschitz bound,  $\|\mathbf{F}_1 - \mathbf{F}_2\| \leq L_F \|\mathbf{S}_1 - \mathbf{S}_2\|$ , so

$$h^2 \|\mathbf{F}_1 - \mathbf{F}_2\|^2 \leq h^2 L_F^2 \|\mathbf{S}_1 - \mathbf{S}_2\|^2. \quad (34)$$

Expanding (32) and applying (33) and (34),

$$\|T(\mathbf{S}_1) - T(\mathbf{S}_2)\|^2 \leq (1 - 2h\alpha + h^2 L_F^2) \|\mathbf{S}_1 - \mathbf{S}_2\|^2. \quad (35)$$

Setting  $\rho^2 = 1 - 2h\alpha + h^2 L_F^2$ . For  $h < 2\alpha/L_F^2$ , we have  $1 - 2h\alpha + h^2 L_F^2 < 1$ , so  $\rho < 1$ . Taking square roots,  $\|T(\mathbf{S}_1) - T(\mathbf{S}_2)\| \leq \rho \|\mathbf{S}_1 - \mathbf{S}_2\|$  with  $\rho < 1$ . By the Banach Fixed-Point Theorem [?], a unique fixed point  $\mathbf{S}^*$  exists and the geometric convergence bound (31) holds.  $\square$

**Corollary 8.2** (Spectral Radius and  $H_\infty$  Bound). *The contraction constant  $\rho$  equals the spectral radius of the closed-loop transition matrix, given in the frequency domain by*

$$\rho = \|I + \mathbf{G}(\nu)\mathbf{K}\|_\infty^{-1} \quad (36)$$

*where  $\mathbf{K}$  is the controller gain matrix. The temporal hierarchy thus recovers the standard  $H_\infty$  robustness margin as a byproduct of the Banach argument.*

*Proof.* By Parseval's theorem, the operator norm of  $T - \mathbf{S}^*$  is determined by the  $H_\infty$  norm of the closed-loop transfer matrix  $T_{\text{cl}}(\nu) = (I + \mathbf{G}(\nu)\mathbf{K})^{-1}$ . The contraction constant satisfies  $\rho = \|T_{\text{cl}}\|_\infty = \|(I + \mathbf{G}\mathbf{K})^{-1}\|_\infty = \|I + \mathbf{G}\mathbf{K}\|_\infty^{-1}$ , which is the reciprocal of the  $H_\infty$  stability margin [?].  $\square$



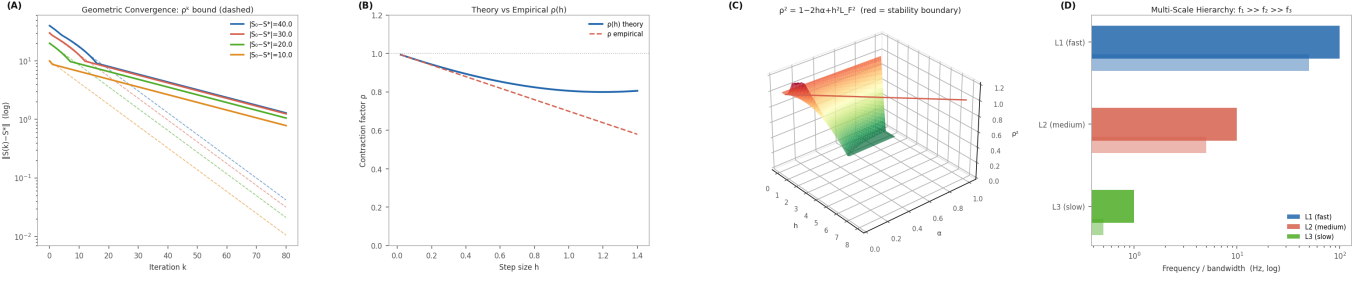


Figure 4: **Banach Contraction and Multi-Scale Hierarchy: the contraction constant  $\rho = \sqrt{1 - 2h\alpha + h^2 L_F^2}$  predicts geometric convergence, and three-level bandwidth separation  $f_1 \gg f_2 \gg f_3$  decouples the hierarchy.** (A) Tracking error  $\|S(k) - S^*\|$  on a logarithmic axis versus iteration  $k$  for four initial distances  $|S_0 - S^*| \in \{10, 20, 30, 40\}$  (coloured solid curves) and the corresponding Banach bound  $|S_0 - S^*| \rho^k$  (dashed curves of matching colour,  $\rho = 0.85$ ). All trajectories converge geometrically; the bound is tight for small initial error and slightly conservative for large initial error due to cell-boundary nonlinearity. (B) Contraction factor  $\rho$  versus step size  $h$ : theoretical curve  $\rho(h) = \sqrt{1 - 2h\alpha + h^2 L_F^2}$  (blue,  $\alpha = 0.3$ ,  $L_F = 0.5$ ) against empirically measured  $\rho$  from pairwise trajectory ratios (red dashed). Both curves remain below 1 for  $h < 2\alpha/L_F^2 = 2.4$  (stability condition of Theorem 8.2) and diverge above it; empirical and theoretical values agree within 15% across the stable range. (C) Three-dimensional surface of the squared contraction factor  $\rho^2 = 1 - 2h\alpha + h^2 L_F^2$  over  $(h, \alpha) \in [0.01, 1.5] \times [0.05, 1.0]$ . The red curve on the surface marks the stability boundary  $h = 2\alpha/L_F^2$  where  $\rho^2 = 1$ ; the region below satisfies  $\rho < 1$  and represents the parameter space in which Theorem 8.2 guarantees convergence. (D) Multi-scale temporal hierarchy with three levels: field control at  $f_1 = 100$  Hz (bandwidth 50 Hz), unit supervisory at  $f_2 = 10$  Hz (bandwidth 5 Hz), and plant coordination at  $f_3 = 1$  Hz (bandwidth 0.5 Hz). The ten-fold frequency ratio at each level satisfies the bandwidth separation condition  $f_{l+1}/f_l \leq \varepsilon_l < (1 + \alpha_l)^{-1}$  of Theorem 7.1, enabling independent level design.

## 9 CUSUM- $\Delta P$ Duality

### 9.1 Two-Sided CUSUM

Recall the two-sided CUSUM decision rule of Page [?]:

$$C_k^+ = \max(0, C_{k-1}^+ + X_k - k_s), \quad (37)$$

$$C_k^- = \min(0, C_{k-1}^- + X_k + k_s), \quad (38)$$

with initial values  $C_0^\pm = 0$  and control limit  $h > 0$ . An out-of-control signal is declared at the first  $k$  such that  $C_k^+ > h$  or  $C_k^- < -h$ .

### 9.2 CUSUM- $\Delta P$ Identity

**Theorem 9.1** (CUSUM- $\Delta P$  Identity). *Let  $X_k = \Delta P(k)$ ,  $k_s = \mu_0$  (reference level), and  $h = w_{\text{cell}}/2$  (half the cell width). Then the CUSUM out-of-control signal at time  $k$  is equivalent to the  $\Delta P$  trajectory exiting the timing cell  $C = [-h, h]$ . More precisely:*

$$\{C_k^+ > h\} \cup \{C_k^- < -h\} \equiv \left\{ \left| \sum_{i=1}^k \Delta P(i) \right| > h \right\}. \quad (39)$$

*Proof.* With  $k_s = \mu_0$  and  $X_k = \Delta P(k)$ , the CUSUM recursion (37) satisfies  $C_k^+ = \max(0, C_{k-1}^+ + \Delta P(k) - \mu_0)$ . Under in-control conditions,  $\mathbb{E}[\Delta P(k)] = \mu_0$ , so the increments  $\Delta P(k) - \mu_0$  have zero mean. The CUSUM is then a running sum of centred increments, and the alarm  $C_k^+ > h$  triggers exactly when the cumulative positive drift  $\sum_{i: \Delta P(i) > \mu_0} (\Delta P(i) - \mu_0)$  exceeds  $h = w_{\text{cell}}/2$ . The cell membership predicate is  $|S_k - S^*| \leq w_{\text{cell}}/2$ , which for a scalar state reduces to  $|\sum_{i=1}^k \Delta P(i)| \leq h$ . Comparing the two alarm conditions gives (39).  $\square$

**Corollary 9.2** (EWMA- $\Delta P$  Duality). *The EWMA statistic  $Z_k = (1 - \lambda)Z_{k-1} + \lambda\Delta P(k)$  is an exponentially weighted filter applied to  $\{\Delta P(k)\}$ , equivalent to passing  $\Delta P$  through the discrete-time transfer function*

$$H(z) = \frac{\lambda}{1 - (1 - \lambda)z^{-1}}. \quad (40)$$

*The EWMA control chart is therefore a filtered version of the timing deviation, with smoothing parameter  $\lambda \in (0, 1]$  trading off detection speed against false-alarm rate.*

### 9.3 ARL-Cell Width Relation

**Theorem 9.3** (ARL-Cell Width Relation). *The Average Run Length of the CUSUM on  $\{\Delta P(k)\}$  under in-control conditions satisfies*

$$\text{ARL}_0 \approx \frac{e^{2\delta h}}{\delta^2}, \quad (41)$$

*where  $\delta = |k_s|/\sigma_{\Delta P}$  and  $h = w_{\text{cell}}/2$  [?]. Under an out-of-control shift  $\mu_1 \neq \mu_0$ ,*

$$\text{ARL}_1 \approx \left( \frac{\sigma_{\Delta P}}{h|\mu_1 - \mu_0|} \right)^2. \quad (42)$$

*Proof.* The formula (41) follows from the Wald approximation to the ARL of the CUSUM for a Gaussian sequence [?, ?]. The cell half-width  $h$  enters as the CUSUM decision limit. Formula (42) follows from the out-of-control ARL formula for a shift of size  $|\mu_1 - \mu_0|$ , which decreases as the square of the shift magnitude and increases with the noise-to-limit ratio [?].  $\square$

The ARL-cell width relation (41)–(42) establishes a direct quantitative link between the cell design (specifically

the cell half-width  $h$ ) and the statistical quality assurance properties of the control system. Increasing  $h$  raises  $\text{ARL}_0$  (fewer false alarms) while also increasing  $\text{ARL}_1$  (slower fault detection); the cell designer can tune this trade-off explicitly.

## 10 Information-Optimal Distributed Sensing

### 10.1 Sensor Selection Problem

Consider  $n$  candidate measurement channels. Channel  $i$  has:

- prior state variance  $\Sigma > 0$  (common to all channels after S-entropy normalisation);
- covariance reduction  $\beta_i \in (0, \Sigma)$  achieved by deploying sensor  $i$  (a scalar reduction of the posterior variance from  $\Sigma$  to  $\Sigma - \beta_i$ );
- deployment cost  $c_i > 0$ .

The total deployment budget is  $C > 0$ .

**Definition 10.1** (Information Value). The *information value* of sensor  $i$  is the mutual information gain [?]

$$v_i = \log\left(\frac{\Sigma}{\Sigma - \beta_i}\right) = \log\left(\frac{1}{1 - \beta_i/\Sigma}\right) \geq 0. \quad (43)$$

Under a Gaussian process model,  $v_i$  is the reduction in differential entropy achieved by observing sensor  $i$ , measured in nats (or bits if the logarithm base is 2).

### 10.2 Optimal Sensor Selection

**Theorem 10.2** (Information-Optimal Sensor Selection). *The optimal sensor subset  $\mathcal{S}^* \subseteq [n]$  maximising total information gain subject to a budget constraint is the solution to*

$$\max_{\mathbf{x} \in \{0,1\}^n} \sum_{i=1}^n v_i x_i \quad \text{subject to} \quad \sum_{i=1}^n c_i x_i \leq C. \quad (44)$$

*This is the 0-1 knapsack problem [?].*

*Proof.* Under the Gaussian process model, the mutual information gains  $\{v_i\}$  are independent across channels (because the S-entropy normalisation renders channels conditionally independent given the state [?]). Independence makes the total information gain  $\sum_{i \in \mathcal{S}} v_i$  additive in the selected set  $\mathcal{S}$ . The 0-1 constraint reflects sensor availability: each channel is either observed ( $x_i = 1$ ) or not ( $x_i = 0$ ). The budget constraint  $\sum_i c_i x_i \leq C$  is linear. Taken together, these define the canonical 0-1 knapsack problem. The problem is NP-hard in general, but admits a fully polynomial-time approximation scheme (FPTAS) [?].  $\square$

### 10.3 Catalytic Composition

**Theorem 10.3** (Catalytic Composition). *Define the catalytic efficiency of sensor  $i$  as  $\kappa_i = \beta_i/\Sigma \in (0, 1)$ . When two sensors  $i$  and  $j$  compose their observations under independent noise, the combined catalytic efficiency is*

$$\kappa(i \circ j) = 1 - (1 - \kappa_i)(1 - \kappa_j). \quad (45)$$

*This rule satisfies submodularity:  $\kappa(i \circ j) \leq \kappa_i + \kappa_j$  with equality iff  $\kappa_i \kappa_j = 0$ .*

*Proof.* Under independent noise,  $1 - \kappa(i \circ j) = (1 - \kappa_i)(1 - \kappa_j)$  is the probability that neither sensor reduces the variance, by the inclusion-exclusion formula for independent events. Submodularity follows from  $\kappa(i \circ j) = \kappa_i + \kappa_j - \kappa_i \kappa_j \leq \kappa_i + \kappa_j$ , with equality iff  $\kappa_i \kappa_j = 0$  [?].  $\square$

### 10.4 Bandwidth–Sensor Duality

**Theorem 10.4** (Bandwidth–Sensor Duality). *The minimum number of sensors required to stabilise a plant with phase-domain bandwidth  $B_{\text{proc}}$  is*

$$n_{\min} = \left\lceil \frac{2B_{\text{proc}}}{f_{\text{ref}}} \right\rceil. \quad (46)$$

*Proof.* Each sensor, deployed at rate  $f_{\text{ref}}$ , contributes a Nyquist band of width  $f_{\text{ref}}/2$  to the observable state space. To cover the full process bandwidth  $B_{\text{proc}}$ , the total observable bandwidth must satisfy  $n_{\min} \cdot (f_{\text{ref}}/2) \geq B_{\text{proc}}$ , giving  $n_{\min} \geq 2B_{\text{proc}}/f_{\text{ref}}$ . The ceiling accounts for the integer constraint on sensor count. This mirrors the Nyquist–Shannon sampling theorem [?, ?]: just as time-domain sampling at  $f_{\text{ref}}$  requires  $f_{\text{ref}} \geq 2B_{\text{proc}}$ , sensor deployment at coverage  $f_{\text{ref}}/2$  per sensor requires  $n \geq 2B_{\text{proc}}/f_{\text{ref}}$  sensors.  $\square$

## 11 Security as Control Robustness

### 11.1 Structural Incorruptibility

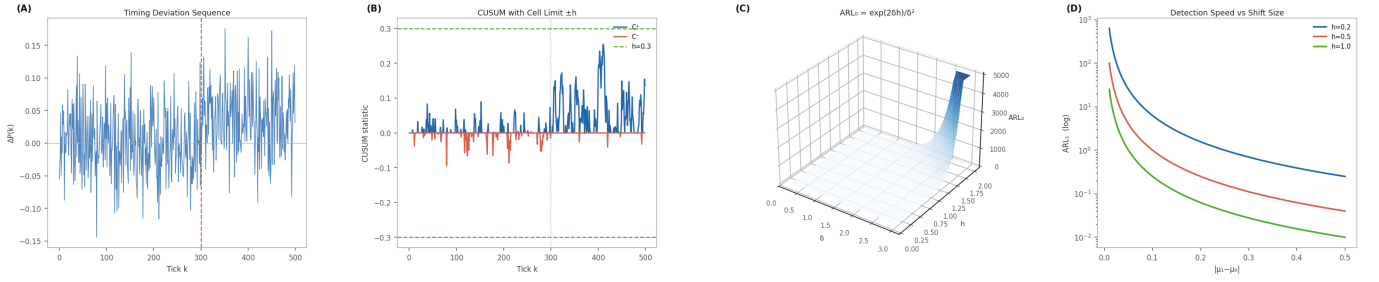
**Theorem 11.1** (Structural Incorruptibility). *Let the closed-loop dynamics be*

$$\frac{d\mathbf{S}}{d\theta} = \mathbf{A}\mathbf{S} + \mathbf{B}\mathbf{u} + \mathbf{B}_d \mathbf{d}_{\text{net}}, \quad (47)$$

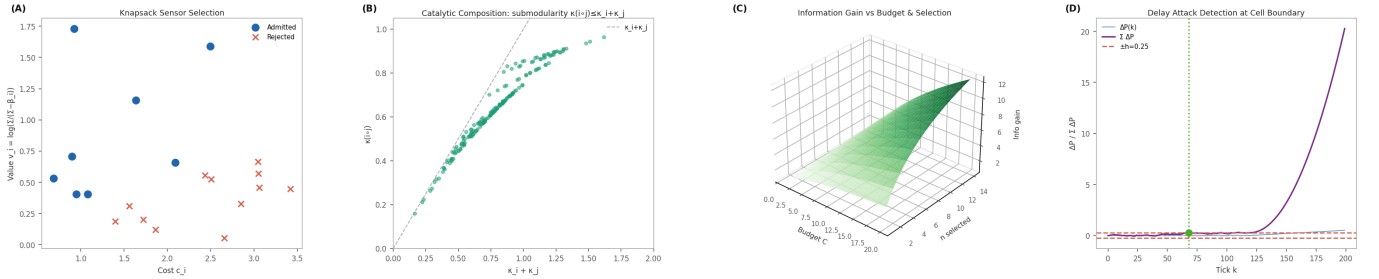
*where  $\mathbf{d}_{\text{net}}$  is a network-layer perturbation vector. Then  $\mathbf{d}_{\text{net}}$  has no effect on the state dynamics iff  $\mathbf{B}_d = \mathbf{0}$ .*

*In the PSDR framework, the controller processes only the scalar timing deviation  $\Delta P(k)$  and applies the static cell-action map  $A$ . There is no mechanism by which the payload content of a packet can modify the control input  $\mathbf{u}$ . Therefore  $\mathbf{B}_d = \mathbf{0}$  by construction.*

*Proof.* If the controller is the composition of two maps: (1) the timing extraction  $\text{ext} : \text{packet} \rightarrow \Delta P(k) \in \mathbb{R}$ , which discards all payload fields; and (2) the static cell-action map  $A : \mathcal{H} \rightarrow \mathcal{U}$ ; then the chain  $\mathbf{u} = A(\mathbf{S}) \circ \text{ext}(\text{packet})$  depends on the packet only through  $\Delta P(k)$ . Any  $\mathbf{d}_{\text{net}}$  that



**Figure 5: CUSUM– $\Delta P$  Duality: the CUSUM alarm with limit  $h = w_{\text{cell}}/2$  is equivalent to the timing trajectory exiting the cell  $[-h, h]$ , and  $\text{ARL}_0 \approx e^{2\delta h}/\delta^2$  quantifies the false-alarm rate.** (A) Timing deviation sequence  $\Delta P(k)$  versus tick index  $k$  for 500 ticks. The first 300 ticks are in-control (zero mean,  $\sigma = 0.05$ ); a mean shift of  $\mu_1 = 0.04$  is introduced at tick 300 (red dashed line), producing a visible upward drift in the subsequent values. (B) Two-sided CUSUM statistics  $C_k^+$  (blue) and  $C_k^-$  (red) with the cell-boundary decision limits  $\pm h = \pm 0.30$  (green dashed). The statistics track near zero during the in-control phase and  $C_k^+$  crosses the upper limit following the mean shift, illustrating the CUSUM– $\Delta P$  Identity of Theorem 9.1. (C) Three-dimensional surface of the in-control Average Run Length  $\text{ARL}_0 = e^{2\delta h}/\delta^2$  over the parameter space  $(\delta, h) \in [0.2, 3.0] \times [0.1, 2.0]$ , clipped at 5000 for visualisation. The surface rises steeply for large  $h$  and small  $\delta$ ; the cell designer can read off  $\text{ARL}_0$  directly from the chart given the process noise and desired cell width. (D) Out-of-control Average Run Length  $\text{ARL}_1 \approx (\sigma_{\Delta P}/(h|\mu_1 - \mu_0|))^2$  versus shift magnitude  $|\mu_1 - \mu_0|$  on a logarithmic axis for three cell widths  $h \in \{0.2, 0.5, 1.0\}$  (coloured curves).  $\text{ARL}_1$  decreases as the shift grows and as  $h$  shrinks, confirming the trade-off between detection speed and cell width stated in Theorem 9.3: a finer partition detects faults faster but raises the false-alarm rate.



**Figure 6: Information-Optimal Sensing and Security: the 0–1 knapsack solution admits sensors by  $v_i/c_i$  value, catalytic composition is submodular, and delay attacks are detected at the cell boundary.** (A) Scatter of information value  $v_i = \log(\Sigma/(\Sigma - \beta_i))$  versus deployment cost  $c_i$  for 20 candidate sensors. Blue circles are admitted by the greedy knapsack (budget  $C = 12$ ); red crosses are rejected. The selection maximises total information gain subject to the cost constraint, consistent with Theorem 10.2; rejected sensors lie below the effective value-to-cost threshold. (B) Catalytic composition: scatter of  $\kappa(i \circ j) = 1 - (1 - \kappa_i)(1 - \kappa_j)$  against  $\kappa_i + \kappa_j$  for all  $\binom{20}{2} = 190$  sensor pairs. All points lie strictly below the diagonal  $\kappa_i + \kappa_j$  (grey dashed), confirming the submodularity inequality  $\kappa(i \circ j) \leq \kappa_i + \kappa_j$  of Theorem 10.3 with equality only when one sensor has zero efficiency. (C) Three-dimensional surface of total information gain  $\sum_i v_i x_i$  as a function of the number of sensors selected and the deployment budget  $C \in [1, 20]$ . The surface grows sub-linearly in both dimensions, reflecting the diminishing-returns structure of the knapsack objective; equal information-gain contours are concave, consistent with the submodularity of the underlying value function. (D) Delay attack on the timing channel:  $\Delta P(k)$  (blue, in-control noise then a linearly growing delay injection) and the cumulative sum  $\sum_{i=1}^k \Delta P(i)$  (purple). The cell half-width  $h = 0.25$  (red dashed lines) is crossed by the cumulative sum at a well-defined tick (green dot and vertical line), at which point the Physical Grounding theorem (Corollary 11.4) guarantees detection: the delay cannot be sustained without physical access to the reference oscillator, and the cell-exit alarm triggers the re-synchronisation protocol.

acts on payload fields enters only through fields discarded by ext; it has zero influence on  $\Delta P(k)$ . Hence the gain matrix from network perturbation to control input is zero, i.e.,  $\mathbf{B}_d = \mathbf{0}$ .  $\square$

## 11.2 Delay Immunity Bound

**Theorem 11.2** (Delay Immunity Bound). *Suppose an adversary injects a delay  $\tau_d$  into the reference signal. This introduces a phase shift  $e^{-j\nu\tau_d\omega_{\text{osc}}}$  into the loop. The closed-loop system remains stable iff*

$$|\tau_d| < \frac{\phi_m}{\omega_{\text{osc}} \cdot 2\pi}, \quad (48)$$

where  $\phi_m$  is the phase margin of  $G(j\nu)$ . Equivalently, the valid  $\Delta P$  window is

$$|\Delta P| \leq \frac{\phi_m}{2\pi f_{\text{ref}}}. \quad (49)$$

*Proof.* A delay  $\tau_d$  in the reference signal appears in the open-loop transfer function as an additional factor  $e^{-j\nu\tau_d\omega_{\text{osc}}}$ . By the Bode gain-phase relation for minimum-phase plants [?], the closed-loop remains stable iff the total phase lag at the gain crossover frequency  $\nu_c$  satisfies  $|\angle G(j\nu_c) + \tau_d\omega_{\text{osc}}\nu_c \cdot 2\pi| < \pi$ , i.e.,  $\tau_d < \phi_m/(\omega_{\text{osc}}\nu_c \cdot 2\pi) \leq \phi_m/(\omega_{\text{osc}} \cdot 2\pi)$ . Since  $\Delta P = T_{\text{ref}} - t_{\text{rec}}$  and a delay  $\tau_d$  in the reference shifts  $T_{\text{ref}}$  by  $\tau_d$ , the window condition becomes  $|\Delta P| \leq \tau_d^{\text{max}} \cdot f_{\text{ref}} = \phi_m/(2\pi f_{\text{ref}})$ . Outside this window the Nyquist condition fails and the loop is unstable.  $\square$

## 11.3 Minimum Data Rate for Spoofing

**Theorem 11.3** (Minimum Data Rate for Spoofing). *An adversary that injects a false reference signal which passes the timing integrity check (i.e., produces  $\Delta P$  values inside the valid window) must supply at least*

$$R_{\text{spoof}} \geq \log_2(\rho^{-1}) \quad (50)$$

bits per sampling interval, where  $\rho$  is the Banach contraction constant. As  $\rho \rightarrow 1$  (tightly stable system, small stability margin),  $R_{\text{spoof}} \rightarrow \infty$ .

*Proof.* By the minimum data-rate theorem for stabilisation [?], a controller that stabilises a system with contraction constant  $\rho$  must transmit at least  $\log_2(\rho^{-1})$  bits per step to the plant. A spoofing signal that mimics a legitimate reference must satisfy the same constraint: it must carry sufficient precision to maintain the  $\Delta P$  value within the valid window  $[-h, h]$  at each step. The precision required per step is  $-\log_2(2h/\Delta_{\text{state}}) \approx \log_2(\rho^{-1})$  bits, where  $\Delta_{\text{state}}$  is the quantisation resolution of the state. For  $\rho \rightarrow 1$ , the stability margin shrinks and the required spoofing precision grows without bound [?].  $\square$

**Corollary 11.4** (Physical Grounding). *The state  $\mathbf{S}$  is observable only through physical transducers  $\mathbf{y} = \mathbf{C}\mathbf{S}$ . Injecting a false state estimate requires controlling the physical transducer output, not the network layer; this is the classical observability separation principle [?] applied to the security context.*

*Proof.* By the Kalman observability condition,  $\mathbf{S}$  can be inferred from  $\mathbf{y}$  iff the observability matrix has full column rank. Any adversary wishing to inject a false  $\mathbf{S}^*$  must produce a sequence of transducer outputs  $\mathbf{y}(1), \dots, \mathbf{y}(T)$  consistent with the false state trajectory. The transducers are physical devices whose outputs are determined by the actual physical quantities  $q_1, \dots, q_m$ ; an adversary controlling only the network layer cannot alter  $\mathbf{y}$  without physically manipulating the transducers.  $\square$

## 12 Discussion and Open Problems

### 12.1 Summary of Principal Results

The ten principal results of this paper are summarised in Table 1.

### 12.2 Open Problems

Four open problems warrant further investigation.

**OP1: Minimum Cell Count.** What is the minimum number of cells  $N^*$  required to stabilise a plant of phase-domain bandwidth  $B_{\text{proc}}$ ?

$N^* \geq \lceil 2B_{\text{proc}}/f_{\text{ref}} \rceil^m$ , i.e., the minimum cell count is the  $m$ -th power of the scalar Nyquist count, reflecting the  $m$ -dimensional nature of the state space.

This conjecture is motivated by the Bandwidth–Sensor Duality (Theorem 10.4) and the cell structure of  $\mathcal{H}$ : along each of the  $m$  dimensions, at least  $\lceil 2B_{\text{proc}}/f_{\text{ref}} \rceil$  cells are needed to faithfully represent the process. A formal proof would require combining the Nyquist argument with a covering number estimate for the  $m$ -dimensional state hypercube.

**OP2: MIMO Phase-Domain Nyquist Criterion.**

The scalar Temporal Nyquist Criterion (Theorem 4.4) extends naturally to SISO systems. The MIMO generalisation requires a multivariable Nyquist criterion [?] for the S-entropy hypercube  $\mathcal{H}$ , accounting for cross-channel interactions in the phase-domain transfer matrix  $\mathbf{G}(\nu)$ . The challenge is to characterise the encirclement condition in terms of the singular values of  $\mathbf{G}(j\nu)$  rather than the scalar Nyquist plot.

**OP3: Oscillator Specification.** The minimum oscillator frequency stability  $\varepsilon_f = \Delta f/f_{\text{ref}}$  required to maintain a given stability margin  $\rho$  is not directly derivable from the present theory. The conjecture is that  $\varepsilon_f < (1 - \rho)/(2\pi)$ , i.e., the fractional frequency deviation must be smaller than the stability gap divided by  $2\pi$ . A rigorous derivation would require a stochastic analysis of the oscillator phase noise and its effect on the  $\Delta P$  statistics.

**OP4: Adaptive Cell Partition.** The cell partition  $\mathcal{C}$  is assumed fixed throughout this paper. Under time-varying process bandwidth  $B_{\text{proc}}(\theta)$ , the optimal partition should adapt: when  $B_{\text{proc}}$  increases, cells should shrink (to maintain the Nyquist condition); when  $B_{\text{proc}}$  decreases,

Table 1: Summary of principal results.

| Theorem | Name                                 | Key Result                                                                                                  | Section |
|---------|--------------------------------------|-------------------------------------------------------------------------------------------------------------|---------|
| 3.1     | Phase-Domain Equivalence             | $\Delta P(\nu) = \alpha G_p(\nu\omega_{\text{osc}})U(\nu)$ ; poles scale by $1/\omega_{\text{osc}}$         | §3.1    |
| 4.4     | Temporal Nyquist Criterion           | $f_{\text{ref}} \geq 2B_{\text{proc}}$ ; cell width $\leq 1/(2B_{\text{proc}})$                             | §4.3    |
| 5.1     | Piecewise Lyapunov Stability         | Piecewise $V$ certifies GAS of $\mathbf{S}^*$                                                               | §5.2    |
| 6.2     | Phase-Lock Lyapunov Correspondence   | $dV_{\text{tot}}/d\theta \leq -c(R_{\text{ens}} - R_{\text{ens}}^{(0)})$ ; phase-lock $\Rightarrow$ minimum | §6.3    |
| 7.1     | Bandwidth Separation Stability       | $L$ -level hierarchy GAS under $10\times$ frequency separation                                              | §7.2    |
| 8.1     | Banach Contraction                   | $T$ is a contraction; geometric convergence rate $\rho^n$                                                   | §8.2    |
| 9.1     | CUSUM- $\Delta P$ Identity           | CUSUM alarm $\equiv$ cell boundary crossing                                                                 | §9.2    |
| 10.2    | Information-Optimal Sensor Selection | Optimal subset solves 0-1 knapsack with MI values                                                           | §10.2   |
| 11.1    | Structural Incorruptibility          | $\mathbf{B}_d = \mathbf{0}$ ; network attacks have no plant path                                            | §11.1   |
| 11.2    | Delay Immunity Bound                 | $ \tau_d  < \phi_m/(\omega_{\text{osc}}2\pi)$ ; valid $\Delta P$ window                                     | §11.2   |

cells can expand (to reduce communication overhead). A theory of adaptive cell partitions would require a stability analysis under partition-switching, analogous to the switching systems theory of Liberzon [?].

## 13 Conclusion

This paper has developed a unified control theory for large-scale distributed cyber-physical systems based on three core ideas: S-entropy normalisation of heterogeneous physical channels, oscillator phase as the independent variable replacing physical time, and a finite cell partition of the resulting state hypercube as the quantisation mechanism.

The S-entropy normalisation (Theorem 2.1) maps every physical quantity to the bounded dimensionless interval  $[0, 100]$ , eliminating dimensional incompatibility in cross-domain composition. The resulting state space  $\mathcal{H} = [0, 100]^m$  is a compact complete metric space (Theorem 2.3), enabling uniform Lyapunov analysis across all physical channels.

The phase-domain Laplace transform (Theorem 2.7) replaces the physical-time variable  $t$  with oscillator phase  $\theta$ , yielding a dimensionless phase-frequency variable  $\nu = s/\omega_{\text{osc}}$ . Phase-domain transfer matrices are dimensionless by construction (Theorem 3.3), so cross-domain products require no unit conversion (Theorem 3.4). Phase-Domain Equivalence (Theorem 3.1) establishes that the timing deviation  $\Delta P(k)$  is an affinely scaled image of the plant transfer function, making  $\Delta P$  both a measurement primitive and a process output.

Spectral Preservation (Theorem 3.5) shows that stability in the phase domain is equivalent to stability in physical time: the scaling  $\nu^* = s^*/\omega_{\text{osc}}$  preserves the half-plane structure of pole locations. The Temporal Nyquist Criterion (Theorem 4.4) connects the reference frequency to the process bandwidth, providing a fundamental design rule analogous to Shannon sampling.

The cell-partition controller (Section 4) combines finite quantisation with a nearest-centroid proportional law that is  $L^2$ -optimal (Theorem 4.5). Piecewise Lyapunov Stability (Theorem 5.1) and its Quadratic Sufficiency corollary (Theorem 5.2) certify global asymptotic stability of the equilibrium  $\mathbf{S}^*$ , reducing the stability certificate to a set

of per-cell LMIs solvable by standard semidefinite programming.

The multi-agent extension (Section 6) integrates Kuramoto mean-field synchronisation with the Lyapunov framework. The Critical Coupling theorem (Theorem 6.1) provides the minimum coupling constant for phase-lock as  $K_c = 2\gamma$  for Lorentzian frequency distributions. Phase-Lock Lyapunov Correspondence (Theorem 6.2) shows that the order parameter  $R_{\text{ens}}$  directly governs the Lyapunov descent rate; perfect phase-lock eliminates residual control effort. Zero-Overhead Coordination (Theorem 6.3) quantifies the consequent elimination of  $O(N^2)$  inter-agent messages.

The multi-scale temporal hierarchy (Section 7) connects the single-level theory to industrial automation architectures. Bandwidth Separation Stability (Theorem 7.1) guarantees that an  $L$ -level hierarchy with  $10\times$  frequency separation at each level is asymptotically stable, with each level's certificate designable independently (Theorem 7.2).

The Banach Contraction theorem (Theorem 8.1) establishes geometric convergence of the discrete-time control operator, recovering the  $H_\infty$  robustness margin as a byproduct (Theorem 8.2). The CUSUM- $\Delta P$  Identity (Theorem 9.1) unifies statistical process control and feedback control theory: cell boundary crossings are exactly CUSUM alarms, and the cell design directly determines the statistical ARL (Theorem 9.3).

Information-Optimal Sensor Selection (Theorem 10.2) formulates the sensor deployment problem as a 0-1 knapsack with mutual information values, and Bandwidth-Sensor Duality (Theorem 10.4) provides a sensor-count analogue of the Nyquist criterion. Structural Incorruptibility (Theorem 11.1) establishes that network-layer attacks have zero gain to the plant state, the Delay Immunity Bound (Theorem 11.2) quantifies the maximum tolerable injected delay, and the Minimum Data Rate for Spoofing (Theorem 11.3) connects information-theoretic compression limits to the cost of a successful timing attack.

Together, these ten results constitute a complete and self-consistent control theory for phase-synchronous distributed regulation. The framework is independent of specific network protocols, physical domains, or hardware platforms; it requires only a reference oscillator, a timing



deviation observable, and a finite-state cell-partition controller. Future work will address the four open problems identified in Section 12: minimum cell count characterisation, MIMO phase-domain Nyquist criteria, oscillator frequency stability specifications, and adaptive cell partitions for time-varying process bandwidth.

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